



Corporate Office Network Security

Secure Network Design & Implementation

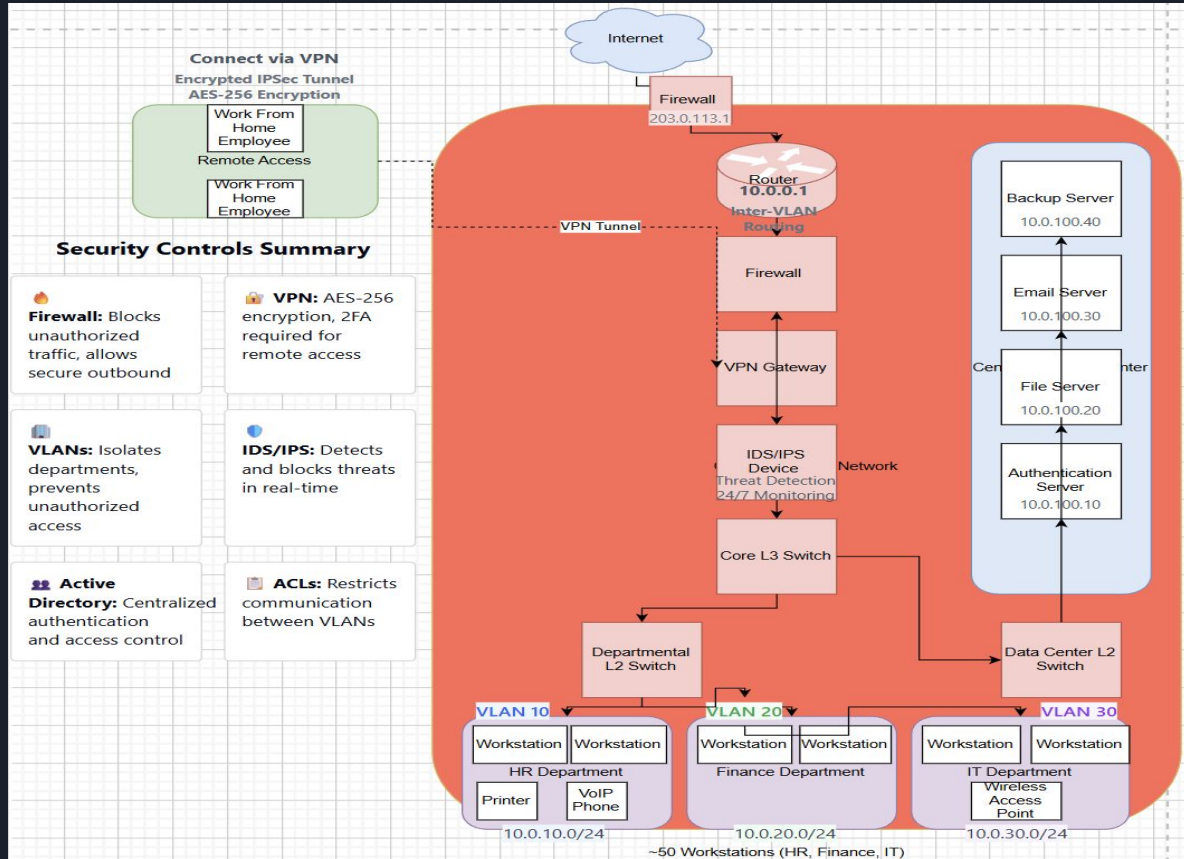


Company Profile

Bullet points:

- Mid-sized company with 150 employees
- Three departments:
 - HR Department (15 employees)
 - Finance Department (20 employees)
 - IT Department (15 employees)
- Challenge: Need secure network with remote access capability
- Goal: Protect sensitive data while enabling productivity

Network Architecture





Department Isolation with VLANs

- **VLAN 10 - HR Department** (10.0.10.0/24)
 - Access to HR files and email only
- **VLAN 20 - Finance Department** (10.0.20.0/24)
 - Access to financial systems with extra security (MFA)
- **VLAN 30 - IT Department** (10.0.30.0/24)
 - Full network management access
- **VLAN 100 - Server Farm** (10.0.100.0/24)
 - Hosts all critical servers

Why? Prevents data leaks between departments



Key Security Devices

- **Firewall:** Blocks malicious traffic from the internet
- **VPN Gateway:** Provides secure remote access with encryption
- **IDS/IPS:** Monitors network for threats and blocks attacks
- **Active Directory:** Controls user authentication and access
- **Layer 3 Switch:** Manages VLAN traffic and routing

Critical Servers (VLAN 100)

Active Directory Server (10.0.100.10)

- Centralized user authentication

File Server (10.0.100.20)

- Shared storage for all departments

Email Server (10.0.100.30)

- Internal and external email

Backup Server (10.0.100.40)

- Daily automated backups for disaster recovery



How We Protect the Network

- **Two-Factor Authentication (2FA)** for VPN and financial systems
- **AES-256 Encryption** for all remote connections
- **Access Control Lists (ACLs)** restrict inter-department communication
- **Daily Backups** with 30-day retention
- **24/7 Monitoring** with IDS/IPS alerts
- **WPA3 Encryption** for wireless networks





Secure Remote Work

Remote workers connect through VPN Gateway

Requirements:

- **Company-approved VPN client**
- **Username + password + 2FA code**
- **Device compliance check**

All traffic encrypted with AES-256

Automatic disconnect after 8 hours for security



Meeting Project Requirements

- ✓ **Prevented internal data leaks** → VLANs isolate departments
- ✓ **Managed access control** → Active Directory + role-based permissions
- ✓ **Secured remote connectivity** → VPN with 2FA and encryption
- ✓ **Protected critical data** → Firewalls, IDS/IPS, and backups



Summary

Successfully designed a secure corporate network

Multi-layered defense protects against internal and external threats

Enables secure remote work for employees

Follows industry best practices (NIST, ISO 27001)

Ready for implementation and regular security updates

This network design ensures confidentiality, integrity, and availability of company resources.